The Great Divide: Finding Things Faster with Binary Search

The Number Guessing Game

Teacher thinks of a number 1-100. Who finds it faster?

Linear: 1, 2, 3, 4... (one by one)

Binary: Start at 50! Higher or lower?

Binary Search divides options in half every time!

What is Linear Search?

Like finding your favourite spice jar in a messy market stall

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Check each item from start to end

Could take many tries if your item is at the end

What is Binary Search?

Like finding a word in a dictionary—jump to the middle!

Requires the list to be sorted (like alphabetical order)

Each guess cuts the search area in half

Why Must The List Be Sorted?

Unsorted names = Binary Search fails!
Sorted roll numbers (1-60) = Perfect!
"Sorting is the secret to Binary Search's speed!"

Step 1: Find the Middle

Look at the middle item in your sorted list.
Start your search right in the middle!

Step 2: Compare With Your Target

Is the middle number your target?
If yes, you're done! 🎉
If not, decide: Is your target higher or lower?

Step 3: Discard Half the List

Cross out the half where your target can't be. You just eliminated half the work!

Step 4: Repeat Until Found

Repeat these steps on the remaining half until you find your target. Watch how quickly the search area shrinks!

Activity Time!

Our school library has 100 books, sorted by ID. Can you find Book #72 using Binary Search?

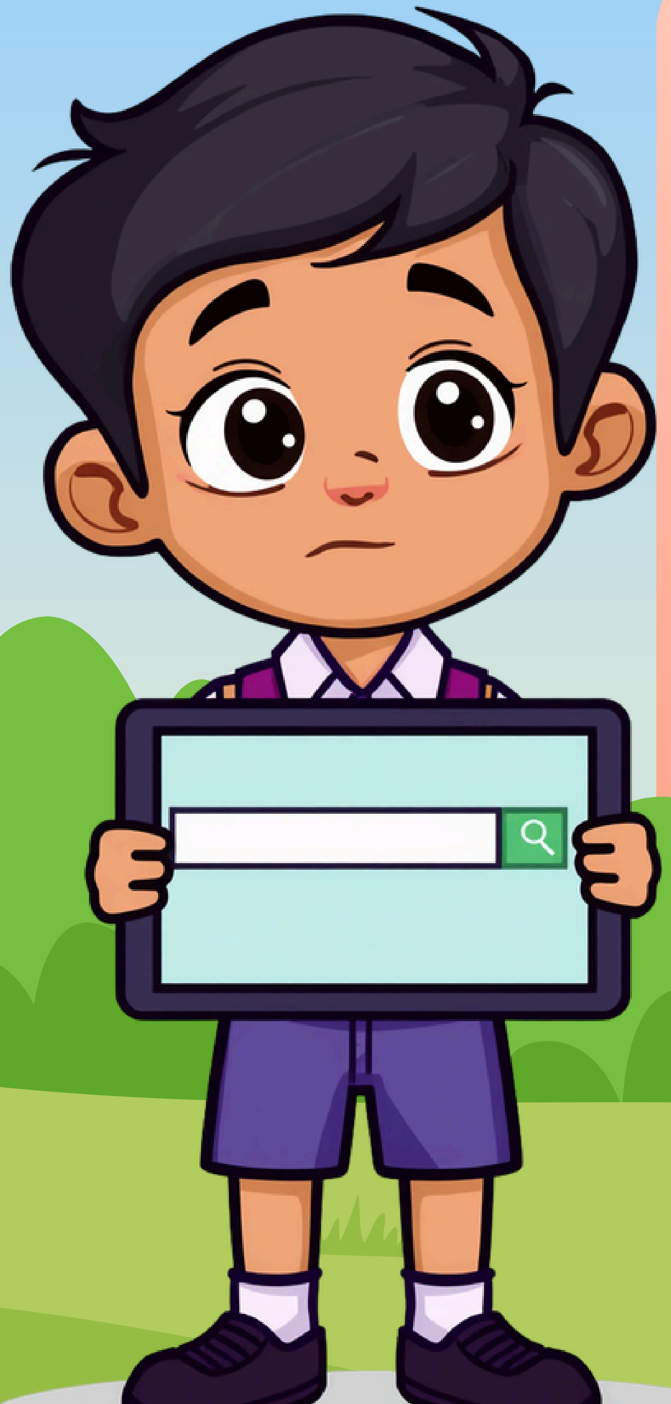
Library Challenge Walkthrough

Guess #50 → Target is higher → Discard 1-50

Guess #75 → Target is lower → Discard 76-100

Continue until Book #72 is found!

Why Does This Matter?



Google finds your search results fast using similar ideas!

IRCTC finds your PNR status from millions quickly!

Algorithms power our everyday life!

Key Takeaways

Linear Search checks one by one; Binary Search divides and conquers

Binary Search only works on sorted lists

Each guess cuts options in half — super fast!

Thank You!